ALAIN HALBOUNY

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Egyptian-French National — UAE Resident — Canada Post Graduate Work Permit

EDUCATION

McGill University - Faculty of Engineering Montreal, Canada, Sep 2020 - Feb 2025
Bachelor of Engineering in Mechanical Engineering GPA: 3.6
Awarded the Professor Bill Williams Memorial and John Howard Ambrose Scholarships 2020 – 2021
Coursework: Thermodynamics, Mechanical Design, Fluid Mechanics, DFMA, GD&T, Numerical Methods, Dynamics, Conceptual Design, Mechanics of Deformable Solids, Electric Machinery, Heat Transfer, Subsonic Aerodynamics

SKILLS

Engineering Tools: CAD: SolidWorks, CAM: Mastercam, FEA: Abaqus
Manufacturing & Design: CNC Machining, 3D Printing, Rapid Prototyping, DFMA, GD&T
Software: Python, MATLAB, Claude (TypeScript, Next.js, React, PostgreSQL, Docker, Git), Higgsfield, Microsoft Office
AI Development: App and website development, workflow automation with AI agents, content creation, AI-assisted CAD
Certificates: Machine Learning Specialization (Stanford), KDB+/q developer level 3 (Kx)
Languages: French, English, Arabic (Fluent); Spanish (Basic)

EXPERIENCE

MEP Engineer - Rowad Modern Engineering Cairo, Egypt, Feb 2026 - Present
Rotating between site and headquarters, supporting senior engineers across disciplines in a fast-paced construction environment.

- Pre-construction of a large indoor multipurpose arena: MEP planning, design coordination, procurement and submittal review.
- Site work across HVAC, fire-fighting, plumbing and electrical: inspections, QC, commissioning and handover.

Aging Material Recovery - 1,400 SKUs sitting idle in the company's main warehouse Aug 2026 - Present

- Recovered EGP 10M, twice the book value, from the first 20% of items through scrap sales, deployment and supplier returns.
- Cleaned up messy ERP inventory data with physical counts and on-site brand checks alongside warehouse staff.
- Initiated QC inspections to decide what to scrap, redeploy or return, then matched stock to site requests.
- Built a multi-user web app with Claude to manage inventory, QC records, decisions, audit trail and dashboards.
- Set up an AI agent workflow to develop and test the app (1,600+ commits in 6 weeks); now integrating live ERP data ahead of company-wide rollout and a mobile app.
- Member of the Supply Chain Committee, working on inventory, logistics and procurement strategy.

Co-Founder - IQASE Oct 2025 - Present
Cofounded a mobile phone accessory startup, taking products from concept to market.

- Designed products and production-ready CAD models; prototyping a new product line.
- Managing manufacturing with Chinese suppliers; mold being cut for the first production run.
- Built the financial and profit models (unit cost, pricing, margins); managing budget and P&L.
- Leading go-to-market: branding, digital marketing and distribution across MENA.
- Leveraging AI (Claude, Lovable, Higgsfield) for marketing, CAD, finance and documentation.

Merchandise Sales Representative - V12 Trackside F1 Store, Formula 1 Grand Prix Montreal, Canada, Jun 2023
GVI-Org - Supervised Voluntary Work Experience Quepos, Costa Rica, Jul 2019
Introductory Programs - ABB for Electrical Industries and GT Advanced Technologies Cairo, Egypt, Jul 2019
Internship – GoPro at ABM Dubai, United Arab Emirates, Feb 2017

PROJECTS

Machine Learning - Personal Project May 2025 - Ongoing
Independently building and optimizing machine learning models for practical applications.

- Implemented a Multi-Layer Perceptron (MLP) from scratch in NumPy to classify handwritten digits using the MNIST dataset.
- Currently developing a small transformer-based language model to explore foundational NLP and attention mechanisms.

Laser Powered Rocket - Team Project Nov 2023 - Apr 2024
Developed a fully functional laser thermal propulsion test apparatus for research applications.

- Designed an aluminum ionization chamber rated for 50 bars, operating with a 350W continuous laser for up to 30s.
- Conducted a first-principles heat transfer study to manage a 15000 K plasma core using heat sinks and regenerative cooling.
- Created production-ready DFMA CAD models, GD&T drawings and assemblies in SolidWorks; led prototyping and machining.
- Managed budget and procurement of outsourced parts, delivering on a tight 6-month timeline and within budget.
- Successfully passed structural integrity tests and produced measurable thrust.

Auxiliary Transmission Box - Team Project Mar - Apr 2022
Designed and optimized a double-reduction auxiliary AGMA standard spur gear transmission box for a small aircraft.

- Conducted detailed AGMA force, stress, and fatigue analysis; engineered shafts, bearings and keys over their lifecycle.
- Applied an iterative design optimization cycle using Python scripts and custom spreadsheets.

Structural Bracket Design - Team Project Sep - Dec 2022
Designed a lightweight 2D acrylic structural bracket to maximize strength and achieve target stiffness.

- Conducted FEA analysis and design optimization in ABAQUS, calibrated with material test data.
- Produced bracket CAD models and DXF files in SolidWorks for laser cutting.

EXTRACURRICULAR | Basketball, Tennis, Padel tennis, Ski, Photography and Video editing, Stock Market, AI